

Questions and Examples

Compiled by : yadavvikas.com

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Basic Level Questions

1. What is LINQ? Why is it used?
2. What are the main advantages of using LINQ in C#?
3. What are the different types of LINQ? Name a few LINQ providers.
4. What is the difference between query syntax and method syntax in LINQ? Provide examples of both.
5. How does deferred execution work in LINQ? Provide an example.
6. What is immediate execution in LINQ? Explain with examples.
7. How does the Where clause work in LINQ? Write a query to filter all even numbers from a list.
8. Explain the purpose of the Select clause in LINQ. How does it differ from SelectMany?
9. What is the difference between First, FirstOrDefault, Single, and SingleOrDefault in LINQ?
10. What is the purpose of the OrderBy and OrderByDescending methods in LINQ?

Intermediate Level Questions

1. What is the difference between IEnumerable and IQueryable in the context of LINQ?
2. Explain the GroupBy operator in LINQ. Provide an example.
3. How can you perform a left join in LINQ? Provide a code snippet.
4. What is the purpose of the Join method in LINQ? Explain its parameters.
5. Write a LINQ query to find the second highest salary in an employee collection.
6. How can you find duplicate values in a collection using LINQ?
7. What is the difference between Take and Skip in LINQ? How can they be used together for pagination?
8. Explain how Aggregate works in LINQ. Write an example to calculate the factorial of a number using Aggregate.
9. How does the Any method differ from the All method in LINQ?
10. What are extension methods in LINQ? Provide examples of commonly used LINQ extension methods.

Advanced Level Questions

1. What is the difference between lazy (deferred) and eager (immediate) execution in LINQ?
2. What are the performance considerations of using LINQ with large datasets? How can you optimize LINQ queries?
3. Explain how LINQ works internally. What is the role of expression trees?
4. How does LINQ handle null values? Demonstrate using a NullReferenceException scenario and its resolution.

5. How can you use LINQ with asynchronous programming (async/await)?
6. Write a LINQ query to group employees by department and count the number of employees in each department.
7. Explain the difference between ToList, ToDictionary, and ToArray in LINQ. When should each be used?
8. What is the Zip operator in LINQ? Provide a use case for combining two sequences.
9. How can LINQ queries be used with Entity Framework? Explain how LINQ-to-SQL differs from LINQ-to-Objects.
10. Write a LINQ query to find the longest string in a list of strings.

Scenario-Based Questions

1. Given a list of students with their grades, write a LINQ query to find the top 3 highest-scoring students.
2. Write a LINQ query to fetch data from two collections: one for products and another for categories, to show product names with their respective categories.
3. Explain how you would implement a LINQ query for hierarchical data (e.g., a company's organizational structure).
4. How would you use LINQ to flatten a nested collection? Provide an example using SelectMany.
5. Given a list of orders and customers, write a LINQ query to find customers who placed no orders.
6. Write a LINQ query to calculate the average, minimum, and maximum of a list of numbers.
7. How would you implement a custom LINQ extension method? Provide an example.
8. Explain how LINQ can be used with XML or JSON data.
9. Write a LINQ query to find employees whose names start with the letter 'A' and sort them by their joining date.
10. Given a dataset with duplicate values, write a LINQ query to remove duplicates.